

Sahil Makim

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Objective

Eager Computer Engineering student specializing in embedded systems, FPGA design, and IoT prototyping. Experienced in low-level firmware, real-time control, and digital design. Seeking a Summer 2026 internship to apply hardware-software co-design skills.

Education

Georgia Institute of Technology | Atlanta, GA

May 2027

Bachelor of Science in Computer Engineering, GPA 3.95

Skills

Programming: C, C++, Verilog, Python, Java, Bash, ARM, RISC-V, MIPS, MATLAB

Hardware: FPGAs, Arm Mbed microcontroller, computer architecture, VHDL, circuit debugging, oscilloscope, logic analyzer

Software: KiCad, SystemVerilog, Visual Studio Code, Altera Quartus II, GitHub, SolidWorks, Linux, Docker, Microsoft Office

Communication: Design proposals, review papers, technical reports, instruction manuals, presentations, Slack,

Experience

Hardware, Security, Trust Division | Georgia Tech Research Institute

Spring 2026

Student Research Assistant

Conducted FPGA software analysis including generated FPGA designs, scripting, and analyzing test results. Coordinated with team members to achieve significant results.

Projects

Nintendo DSi Homebrew Application | Undergraduate Research

Fall 2025

Lead Developer

Developed a homebrew Nintendo DSi game as part of undergraduate research. Team-based project.

- Directed a 4-member team to produce a functional Pong game with music, adjustable speeds, and full DS hardware integration.
- Engineered touchscreen, button, and wireless I/O using DevkitPro SDK and DS-specific libraries.
- Orchestrated architecture design, low-level debugging, and performance validation to ensure reliable operation.

Embedded System Info Center | Personal Project

Summer 2025

Built a modular embedded monitoring system to collect and display environmental sensor data. Individual project.

- Programmed Arduino Uno (C/C++) to interface with 25 sensors/modules, streamlining data collection and responsiveness.
- Devised a dynamic LCD interface that visualized real-time temperature and light changes.
- Assembled and wired all components, assigning I/O pins and synchronizing sensor outputs for accurate display results.

MBED Wordle Game Design | Programming for Hardware/Software Systems

Fall 2024

Replicated the Wordle gameplay experience on limited Mbed microcontroller hardware. Individual project.

- Assigned user interaction through pushbuttons and navigation switches using Mbed OS for responsive input control.
- Expanded the original game by integrating animations, difficulty modes, and scoring systems to boost replay value.

Servo Motor Peripheral Design for FPGA | Digital Design Laboratory

Spring 2025

Developed a servo motor control peripheral on FPGA hardware integrating Verilog and ARM assembly. Team-based project.

- Constructed PWM-based servo control achieving $\pm 2^\circ$ precision for positional control and variable-speed operation.
- Integrated separate ARM control code to interface with FPGA hardware through peripheral communication.
- Verified proper signal propagation using oscilloscopes and logic analyzers to ensure precise signal timing.

Relevant Coursework

Architecture, Systems and Concurrency: pipelining, cache/memory hierarchy, I/O, OS interaction, RISC-V/ARM concepts.

Design & Analysis of Algorithms: Basic techniques of design and analysis of efficient algorithms for standard computational problems. NP-Completeness.

Digital Design Laboratory: Design and implementation of digital systems, including a team design project. CAD tools, project design methodologies, logic synthesis, and assembly language programming.

Programming for Hardware/Software Systems: Creation of execution and storage mechanisms, based on instruction set architecture, for software design including high-level programming languages and operating systems.

Fundamentals of Digital Design: Boolean algebra, gates, flip-flops, data path, memory, RTL/Verilog.

Data Structs & Algorithms: Computer data structures and algorithms in the context of object-oriented programming. Focus on software development towards applications.